Stock Market Analysis in SQL

Final Insights Report — Corrected Version

Analysis period: 1 January 2015 to 31 July 2018 | Stocks: Bajaj Auto, Eicher Motors, Hero Motocorp, Infosys, TCS and TVS Motors

1. Executive Summary

The analysis examined historical closing prices, 20-day and 50-day moving averages, Golden Cross Buy/Sell/Hold signals, daily price movements, unusual price cliffs and historical price adjustments across six NSE stocks. TVS Motors recorded the highest unadjusted first-to-last price increase at 86.9%, followed by Eicher Motors at 82.6%. TCS and Infosys initially appeared to decline, but their approximately 50% one-day price cliffs were investigated and the historical series was adjusted. After adjustment, TCS showed +52.4% and Infosys +38.2% over the period.

2. Stock Performance

Stock	First Close	Last Close	Unadjusted %	Adjusted % where applicable
TVS Motors	276.85	517.45	+86.9%	—
Eicher Motors	15,239.15	27,820.95	+82.6%	—
Bajaj Auto	2,454.10	2,700.70	+10.0%	—
Hero Motocorp	3,107.30	3,293.80	+6.0%	—
TCS	2,548.20	1,941.25	-23.8%	+52.4%
Infosys	1,975.80	1,365.00	-30.9%	+38.2%

Interpretation: TVS Motors had the strongest unadjusted growth. The negative raw returns for TCS and Infosys should not be treated as final historical-performance conclusions because the two series contain large price-level changes.

3. Buy / Sell / Hold Signal Analysis

Stock	Buy	Sell	Hold	Latest non-Hold
Bajaj Auto	12	11	866	Buy
Eicher Motors	6	7	876	Sell
Hero Motocorp	9	9	871	Sell
Infosys	9	9	871	Buy
TCS	12	13	864	Sell
TVS Motors	8	8	873	Sell

Across all six stocks there were 56 Buy signals, 57 Sell signals and 5,221 Hold days. The latest non-Hold signal was Buy for Bajaj Auto and Infosys, and Sell for Eicher Motors, Hero Motocorp, TCS and TVS Motors.

4. Latest Signal vs. Long-Run Trend

Bajaj Auto (+10.0%) and Infosys (+38.2% adjusted) had positive historical performance and a latest Buy signal. Eicher Motors (+82.6%), Hero Motocorp (+6.0%), TCS (+52.4% adjusted) and TVS Motors (+86.9%) had positive adjusted/unadjusted first-to-last performance but a latest Sell signal. This illustrates that a moving-average crossover describes the recent relationship between two averages, while first-to-last performance describes the entire analysis period.

5. Worst Trading Days and the Corrected Infosys Event

Stock	Worst Day	Daily Move
TCS	2018-05-31	-50.4%
Infosys	2015-06-15	-49.9%
TVS Motors	2016-05-03	-9.7%
Eicher Motors	2015-08-24	-9.3%
Bajaj Auto	2015-08-24	-9.1%
Hero Motocorp	2018-07-23	-6.2%

> **Corrected Infosys event:** the price-level change occurred on 2015-06-15. Infosys closed at 1,976.65 on 2015-06-12 and 991.10 on 2015-06-15. The investigation window was therefore corrected to 2015-06-11 through 2015-06-17.

TCS moved from 3,517.75 on 2018-05-30 to 1,744.80 on 2018-05-31. Both events are approximately 50% price-level changes and require historical adjustment before interpreting long-run returns.

6. Effect of Historical Price Adjustment

Stock	Raw % Change	Adjusted % Change
TCS	-23.8%	+52.4%
Infosys	-30.9%	+38.2%

After adjustment, both TCS and Infosys become positive performers. This demonstrates why corporate-action price-level effects must be investigated before comparing historical returns or building trend indicators from raw prices.

7. Signal Flipping / Whipsaw Risk — Corrected Interpretation

The analysis identified 32 Buy/Sell reversals occurring within 30 days. For a normal long-only interpretation, the relevant round trip is Buy → Sell; Sell → Buy should not be treated as a short position unless a short-selling strategy has explicitly been defined.

> **Long-only result:** there were 10 Buy → Sell reversals within 30 days; 9 of the 10 were loss-making, with an average gross price return of approximately -5.05%. These are gross price movements, not realised P&L;, because brokerage, taxes, slippage and position sizing are unavailable.

The 32-reversal count therefore demonstrates potential whipsaw exposure, while the 10 Buy → Sell cases provide the appropriate long-only loss perspective.

8. TCS Latest Signal Caveat

The TCS latest raw-price signal was Sell on 2018-06-05. That signal occurs immediately after the large 2018-05-31 price cliff, so it is strongly influenced by the unadjusted price series. It should not be treated as a clean fundamental or adjusted-series trend conclusion. Rebuilding TCS and Infosys MA20/MA50 signals on the adjusted series is an optional stretch analysis and is not part of the core adjusted-return calculation.

9. What Changed After the Corrections?

The main corrections do not change the core checkpoint numbers. They improve the correctness and presentation of the project: the Infosys event date is corrected to 2015-06-15; the Infosys investigation window is corrected; the final six-stock summary follows the guide's five-column output; the worst-day output follows the guide's four-column format and one-decimal rounding; the first-vs-last output follows the guide's four-column format; and the whipsaw section uses a long-only Buy → Sell interpretation.

10. Limitations

• The analysis uses historical closing prices and does not include dividends.

• Brokerage charges, taxes and slippage are not available, so exact trading P&L; cannot be calculated.

• The analysis does not model company fundamentals, earnings, news or broader market/sector effects.

• The Buy/Sell/Hold rule is a historical analytical rule, not an investment recommendation.

• TCS and Infosys adjusted returns correct the identified price-level effect but do not create a complete total-return series.

• Adjusted-price MA/signal rebuilding for TCS and Infosys is an optional stretch analysis.

11. Final Conclusion

The SQL project successfully combines six stock histories, validates data quality, calculates moving averages, generates crossover signals, evaluates historical performance, identifies extreme daily movements and corrects two important price-level effects. The corrected analysis shows that TVS Motors and Eicher Motors had the strongest unadjusted growth, while TCS and Infosys demonstrate why raw historical prices can be misleading around corporate actions. The crossover strategy generated relatively few signals and showed measurable whipsaw risk. Overall, the project demonstrates an important analyst workflow: calculate, validate, investigate anomalies, correct the data where justified, and only then draw business conclusions.